

NOVATECH SOLUTIONS

HR Attrition Analytics Dashboard

Business Requirements Document

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Prepared as a portfolio simulation using a fictional company and synthetic workforce data.

Document Control

Field	Details
Document Title	Business Requirements Document - HR Attrition Analytics Dashboard
Project Sponsor	HR Director (Fictional Stakeholder)
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Revision History

Version	Date	Author	Change Summary
0.1	July 2026	Amirreza Akbari	Initial business requirements draft
0.9	July 2026	Amirreza Akbari	Requirements aligned with completed workbook and validated KPIs
1.0	July 2026	Amirreza Akbari	Final portfolio release

Distribution and Intended Use

This document is intended for portfolio reviewers, recruiters, hiring managers, and technical stakeholders who want to understand the business context and requirements behind the completed HR attrition analytics solution. The company and dataset are fictional and synthetic; no real employee information is included.

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1. Document Purpose

This Business Requirements Document defines the business need, objectives, scope, stakeholder expectations, analytical requirements, and acceptance criteria for the NovaTech HR Attrition Analytics Dashboard. It provides a common reference for the fictional HR sponsor, business users, and the data analyst responsible for delivery.

Purpose of the solution

Transform employee-level HR data into a transparent, decision-oriented Excel dashboard that identifies where attrition is concentrated, highlights the strongest observed associations, and supports targeted retention actions.

2. Executive Summary

Employee attrition can increase recruitment cost, disrupt service delivery, reduce team productivity, and weaken organizational knowledge. NovaTech requires a repeatable method for monitoring workforce turnover and identifying employee groups that warrant further investigation.

The completed analytical solution uses Microsoft Excel to consolidate, clean, validate, summarize, and visualize workforce information. The final analysis contains 1,468 unique employee records, including 226 recorded departures and 1,242 active employees, resulting in an overall attrition rate of 15.4%.

The dashboard evaluates attrition by department, job role, overtime, age group, tenure group, job satisfaction, income group, gender, and marital status. It is designed for descriptive decision support. Findings indicate associations and prioritization opportunities; they do not establish causality.

Key business need

HR leadership needs to distinguish between high attrition rates caused by small populations and high-volume workforce problems that can materially affect operations.

3. Business Background

NovaTech Solutions is a fictional medium-sized organization used as the business setting for this portfolio case study. The organization employs people across Research and Development, Sales, Human Resources, and a small number of records with an unrecorded department.

The HR function maintains employee-level information covering demographics, organizational assignment, compensation, job conditions, satisfaction, experience, and attrition status. Prior reporting relied on raw spreadsheets and did not provide a consolidated executive view of turnover risk.

The business requires an accessible solution that can be opened and reviewed without specialist software. Microsoft Excel was selected because it supports transparent calculations, PivotTables, slicers, charts, and distribution to business users.

4. Current-State Assessment

Current-State Issue	Business Consequence	Required Improvement
Raw and cleaned data are separated but not summarized for executives	Decision-makers must interpret employee-level records manually	Provide validated KPIs and executive visuals
Department and job-role populations differ materially	Counts alone can misrepresent risk	Present attrition rates together with employee volume
Some categories have small sample sizes	High percentages may be unstable	Include explicit small-sample cautions
Nineteen department values are unrecorded	Department reporting is incomplete	Retain and label as Unknown Department; disclose limitation
No recurring reporting process exists	Insights may become outdated	Design the workbook for repeatable refresh and periodic review

5. Problem Statement

NovaTech lacks a centralized analytical dashboard that allows HR leaders to monitor workforce attrition, compare employee segments, and prioritize retention initiatives using consistent definitions and validated calculations.

Without a structured solution, decision-makers may focus on raw departure counts, overreact to small groups with high percentages, or overlook actionable relationships such as overtime, early tenure, and satisfaction. The project must therefore provide both quantitative evidence and interpretation guidance.

6. Business Objectives

6.1 Primary Objective

Develop an executive-ready Excel dashboard that measures workforce attrition and identifies the employee segments and workplace conditions most strongly associated with turnover.

6.2 Secondary Objectives

- Create a reliable and auditable analytical dataset while preserving the original raw data.
- Calculate and validate core workforce KPIs.
- Compare attrition rates across departments and job roles.
- Evaluate overtime, age, tenure, satisfaction, income, gender, and marital status.
- Balance percentage-based findings with employee population and departure volume.
- Produce evidence-based recommendations for HR leadership.
- Document data limitations and avoid unsupported causal claims.
- Publish the finished solution as a reproducible portfolio project.

7. Business Questions

1. What is the overall attrition rate and how many employees have left the organization?
2. Which departments have the highest attrition rates and departure volumes?
3. Which job roles present the greatest operational retention risk?
4. How does overtime status relate to employee attrition?
5. Which age groups and tenure groups experience the highest attrition?
6. How does job satisfaction relate to retention?
7. Do income bands show meaningful differences in attrition?
8. Are gender and marital status useful attrition differentiators?
9. Which results require small-sample caution?
10. What actions should HR prioritize based on the descriptive findings?

8. Project Scope

8.1 In Scope

- Import and preservation of the original HR dataset.
- Data cleaning, normalization, duplicate handling, and quality documentation.
- Creation of analytical helper fields and employee groupings.
- KPI calculation and reconciliation to PivotTables.
- Attrition analysis by department, job role, overtime, age, tenure, satisfaction, income, gender, and marital status.
- Executive Excel dashboard, insights sheet, recommendations sheet, and supporting documentation.
- Publication of workbook, datasets, dashboard image, reports, and README in GitHub.

8.2 Out of Scope

- Predictive machine-learning models or individual employee risk scoring.
- Causal inference or claims that a single variable causes attrition.
- Live HRIS, payroll, or recruitment-system integration.
- Automated scheduled refresh or email distribution.
- Voluntary versus involuntary attrition segmentation.
- External market-pay benchmarking.
- Use of confidential or real employee records.

9. Stakeholders and Responsibilities

Stakeholder	Role	Primary Responsibility	Decision / Usage
HR Director	Project Sponsor	Defines the business need and approves priorities	Uses findings to prioritize retention initiatives
HR Manager	Primary Business User	Reviews workforce patterns and supports investigation	Monitors departments, roles, and employee segments
Executive Leadership	Decision Maker	Reviews headline KPIs and recommended actions	Approves organizational interventions
Department Managers	Operational Stakeholders	Provide business context for workload and satisfaction findings	Implement team-level actions
Data Analyst - Amirreza Akbari	Project Owner	Cleans data, validates calculations, builds dashboard, interprets findings, documents limitations	Maintains analytical logic and deliverables
Portfolio Reviewer	External Reviewer	Assesses analytical, Excel, and communication capabilities	Evaluates project quality and reproducibility

9.1 Responsibility Matrix

Activity	HR Director	HR Manager	Data Analyst	Executive Team
Define business objectives	A	R	C	C
Validate data definitions	C	R	A/R	I
Develop analytical workbook	I	C	A/R	I
Validate KPIs and findings	C	R	A/R	I
Approve recommendations	A	R	C	C
Use dashboard for decisions	C	R	I	A/R

R = Responsible, A = Accountable, C = Consulted, I = Informed.

10. Functional Requirements

ID	Requirement	Priority	Acceptance Evidence
FR-01	Preserve the original source data in an unchanged Raw_Data worksheet.	High	Raw_Data sheet remains available and auditable.

ID	Requirement	Priority	Acceptance Evidence
FR-02	Provide an analysis-ready Cleaned_Data worksheet with one valid record per employee.	High	1,468 unique employee records are available for analysis.
FR-03	Calculate total employees, departures, active employees, attrition rate, average age, average monthly income, and average tenure.	High	KPI values reconcile with Pivot_Tables.
FR-04	Compare attrition by department using employee count, departure count, and attrition rate.	High	Department PivotTable and dashboard visual are available.
FR-05	Compare attrition by job role and distinguish high-rate/low-volume groups from high-volume risks.	High	Role analysis includes rate and population.
FR-06	Analyze overtime status and its association with attrition.	High	Overtime rates are visible for Yes and No groups.
FR-07	Analyze age, tenure, satisfaction, and income groups.	High	Group PivotTables and visuals are available.
FR-08	Analyze gender and marital status to identify weak or strong differentiation.	Medium	Demographic PivotTables are documented.
FR-09	Allow users to change the displayed measure or filter relevant workforce categories.	Medium	Dashboard controls respond without calculation errors.
FR-10	Provide written insights linked to validated calculations.	High	Insights sheet contains quantified findings.
FR-11	Provide business recommendations directly connected to the findings.	High	Recommendations sheet contains prioritized actions.
FR-12	Disclose data-quality limitations and small-sample risks.	High	Unknown Department and small populations are explicitly noted.

11. Non-Functional Requirements

ID	Category	Requirement	Measurement / Standard
NFR-01	Accuracy	All dashboard KPIs must reconcile to the cleaned dataset and PivotTables.	No unexplained differences between dashboard and supporting calculations.
NFR-02	Usability	A nontechnical HR stakeholder should understand the headline findings within approximately 30 seconds.	Clear title, KPI cards, readable labels, limited color palette.
NFR-03	Performance	Dashboard interactions must	Slicers and measure controls

ID	Category	Requirement	Measurement / Standard
		remain responsive in Microsoft Excel.	update without excessive delay.
NFR-04	Maintainability	Raw, cleaned, analytical, and presentation layers must remain separated.	Dedicated worksheets and documented field definitions.
NFR-05	Transparency	Calculations and limitations must be visible and explainable.	Pivot_Tables, Data_Dictionary, Insights, and Recommendations included.
NFR-06	Portability	The workbook should open in a current desktop version of Microsoft Excel.	No dependence on external add-ins or inaccessible links.
NFR-07	Privacy	Only synthetic employee data may be published.	No real names, contact details, or confidential records.
NFR-08	Visual Quality	The dashboard should use consistent typography, spacing, colors, and number formats.	No clipped labels, misleading scales, or 3D charts.

12. Data Requirements

12.1 Dataset Characteristics

Metric	Requirement / Final Value
Raw source records	1,488 employee records plus header
Final analytical records	1,468 unique employees
Original source fields	35
Analytical fields	Additional helper columns created in Cleaned_Data
Target field	Attrition (Yes / No)
Primary identifier	EmployeeNumber
Data source	Synthetic HR employee dataset
Reporting type	Employee snapshot / descriptive analysis

12.2 Data-Quality Rules

- EmployeeNumber must be unique in the final analytical dataset.
- Attrition must contain only valid Yes or No values.
- Age, income, satisfaction, and tenure fields must be numeric and within documented ranges.
- Department labels must be standardized; missing department values must be retained as Unknown Department.
- Helper group definitions must be consistent across all records.
- Raw data must remain unchanged for auditability.
- All published data must be synthetic and suitable for public portfolio use.

13. KPI Definitions

KPI	Definition	Final Value	Business Interpretation
Total Employees	Count of unique employee records in Cleaned_Data	1,468	Population included in the analysis
Departures	Count of employees where Attrition = Yes	226	Employees recorded as having left
Active Employees	Count of employees where Attrition = No	1,242	Employees remaining in the workforce snapshot
Attrition Rate	Departures divided by Total Employees	15.4%	Approximately one departure for every 6.5 employees
Average Age	Mean employee age	36.4 years	Headline workforce demographic indicator
Average Monthly Income	Mean MonthlyIncome	9,441	Headline compensation indicator; currency follows source assumptions
Average Tenure	Mean YearsAtCompany	3 years, 8 months	Headline employee-service indicator

13.1 Required Segment Metrics

Every major segmentation must present both the attrition rate and the underlying employee population. Where useful, the number of departures must also be shown. This requirement reduces the risk of prioritizing a high percentage that is based on a very small group.

14. Assumptions and Constraints

14.1 Assumptions

- The source Attrition field accurately represents employee departure status.
- Each valid EmployeeNumber represents one employee.
- Satisfaction scales and coded values are consistently applied.
- MonthlyIncome is comparable across the employee population.
- The snapshot is sufficient for descriptive workforce analysis.
- Synthetic data is appropriate for public portfolio demonstration.

14.2 Constraints

- The project is implemented in Microsoft Excel and does not use a live data connection.
- The data does not include a multi-period attrition history.
- No manager, team, exit-interview, absence, promotion-history, or market-pay variables are available.
- Small samples limit confidence in certain role and income findings.
- Descriptive comparisons cannot prove causal relationships.

15. Risks and Mitigations

Risk ID	Risk	Likelihood	Impact	Mitigation
R-01	Duplicate or invalid employee records distort	Medium	High	Validate EmployeeNumber and

Risk ID	Risk	Likelihood	Impact	Mitigation
	KPIs.			document removed records.
R-02	Missing department values create incomplete organizational reporting.	High	Medium	Retain as Unknown Department and disclose the limitation.
R-03	High rates in small groups are over-interpreted.	High	High	Display sample size and distinguish rate from business volume.
R-04	Stakeholders interpret association as causation.	Medium	High	Use careful language and include analytical limitations.
R-05	Dashboard figures do not match PivotTables.	Low	High	Perform KPI reconciliation and validation checks.
R-06	Published workbook contains personal or confidential data.	Low	High	Use synthetic data only and inspect files before publication.
R-07	Dashboard becomes outdated when data changes.	Medium	Medium	Recommend monthly or quarterly refresh and insight updates.

16. Success Criteria and Acceptance

Criterion	Acceptance Standard	Status
Data preparation	Cleaned_Data contains 1,468 unique analysis-ready employees.	Met
KPI accuracy	Headcount, departures, active employees, and attrition rate reconcile to Pivot_Tables.	Met
Analytical coverage	All required employee segments are analyzed.	Met
Dashboard usability	Headline workforce position and risk areas are understandable without opening source data.	Met
Insight quality	Findings include rates, counts, and small-sample caution where necessary.	Met
Recommendation quality	Actions are linked to observed findings and avoid unsupported causal claims.	Met
Documentation	Workbook includes Data_Dictionary, Insights, and Recommendations; repository documentation is prepared.	In progress
Reproducibility	Repository contains workbook, datasets, screenshot, documentation, and	In progress

Criterion	Acceptance Standard	Status
	README.	

17. Project Deliverables

Deliverable	Description	Format	Status
Raw dataset	Original synthetic HR dataset preserved without analytical changes	CSV / worksheet	Completed
Cleaned dataset	Validated employee-level analytical dataset	CSV / worksheet	Completed
Excel analytical workbook	Raw data, cleaned data, dictionary, pivots, dashboard, insights, recommendations	XLSX	Completed
Dashboard image	Publication-ready dashboard preview	PNG	Completed
Business Requirements Document	Business need, scope, requirements, risks, and acceptance criteria	PDF	Completed
Data Dictionary	Expanded dataset definitions and coded-value documentation	XLSX / PDF	Planned
Project Methodology	Technical and analytical workflow documentation	PDF	Planned
Executive Report	Concise management findings and recommendations	PDF	Planned
Portfolio Case Study	Recruiter-oriented end-to-end project narrative	PDF	Planned
Validation Checklist	Quality-assurance evidence	PDF	Planned
GitHub README	Public landing page and project navigation	Markdown	Planned

18. Implementation Timeline

Phase	Key Activities	Output	Status
1. Business understanding	Define problem, questions, scope, and expected value	Business context and requirements	Completed
2. Data preparation	Inspect, clean, standardize, and validate records	Cleaned_Data	Completed
3. Feature engineering	Create attrition flag, age, income, tenure, and satisfaction groupings	Analytical helper fields	Completed

Phase	Key Activities	Output	Status
4. Analysis	Build PivotTables and reconcile rates and counts	Validated analysis tables	Completed
5. Dashboard	Develop KPI cards, charts, controls, and layout	Executive dashboard	Completed
6. Interpretation	Write quantified insights and recommendations	Insights and Recommendations sheets	Completed
7. Publication	Prepare PDF documentation, datasets, screenshot, and README	GitHub portfolio package	In progress

19. Out of Scope and Future Opportunities

The current project is intentionally limited to descriptive Excel analysis. Future development may include:

- Power BI dashboard with automated refresh and drill-through analysis.
- SQL database model for employee, department, role, performance, and turnover data.
- Longitudinal cohort analysis of hiring, promotion, and departure events.
- Voluntary versus involuntary attrition analysis.
- Role-adjusted compensation benchmarking.
- Manager- and team-level retention analysis.
- Predictive modeling using Python, with appropriate governance and fairness controls.
- Measurement of retention interventions over time.

20. Approval and Sign-Off

Because this is a portfolio simulation, the approval section documents the intended governance process rather than a real corporate authorization.

Role	Name	Approval / Signature	Date
Project Sponsor - HR Director	Fictional Stakeholder	_____	_____
Business User - HR Manager	Fictional Stakeholder	_____	_____
Project Owner - Data Analyst	Amirreza Akbari	_____	July 2026

Appendix A - Validated Headline Findings

Analytical Area	Validated Finding	Interpretation Note
Overall	15.4% attrition; 226 of 1,468 employees	Approximately one in 6.5 employees
Department	Sales: 19.6% (87 of 443)	Highest departmental rate and meaningful departure volume
Job role	HR Business Partner: 23.8% (5 of 21)	Highest rate but small sample
Job role - volume	Sales Representative: 21.7% (33 of 152)	High rate with meaningful population

Analytical Area	Validated Finding	Interpretation Note
Overtime	Yes: 23.6%; No: 12.3%	Strongest observed association in the descriptive analysis
Age	Under 25: 26.3%	Highest age-group attrition
Tenure	0-1 years: 21.3%; 10+ years: 10.8%	Indicates an early-tenure retention gap
Job satisfaction	Low: 20.2%; Very High: 12.9%	Attrition generally declines as satisfaction improves
Income	Low Income: 29.4% (n=17)	Small sample; monitor as a hypothesis
Data quality	19 employees have Unknown Department	Known limitation, not a separate business unit

Appendix B - Glossary

Term	Definition
Attrition	Employee departure from the workforce snapshot, recorded as Yes in the source data.
Attrition Rate	Number of departures divided by the employee population for the relevant group.
Headcount	Number of unique employees included in the analysis.
KPI	Key performance indicator used to summarize workforce performance or risk.
PivotTable	Excel summary structure used to aggregate employees, departures, and rates.
Synthetic Data	Artificially generated or simulated data that does not represent real individuals.
Descriptive Analysis	Analysis that summarizes observed patterns without proving causal relationships.
Small-Sample Caution	A warning that a high percentage based on few employees may be unstable or misleading.